

CARES Lab Demonstration

A guided, multi-robot showcase in which several research robots introduce themselves and their projects to visitors, coordinated by a central server.

1 · Project Description

CARES (Centre for Automation and Robotic Engineering Science) researches intelligent robots that can communicate, collaborate, and assist people in real-world environments. This demonstration presents that work as a short, hosted tour: a guide robot welcomes visitors, introduces the lab, then hands off to each research robot in turn. Each robot greets the visitors, explains its own project in accessible language, and answers questions live before the guide moves the group on to the next project.

The robots

Pepper — guide/host	Welcomes visitors, introduces the lab and each project, runs the Q&A windows, and links the demo together.
ChatBox — Project A	Conversational AI & retrieval-augmented generation (RAG): long, context-aware conversation backed by a searchable knowledge base.
Navel — Project B	Emotion-aware interaction: recognising facial expression and tone of voice and adapting the robot's communication style in real time.
Silbot — Project C	Human-aware navigation: moving through shared, crowded spaces safely, politely, and predictably — the way a person would.

How it works

- **Inverted architecture.** Each robot runs a WebSocket server and waits; the central server dials out to it. Robots auto-register with the server (auto-detecting their own IP) and can start automatically on boot.
- **Orchestrated sequence.** A *DemoOrchestrator* state machine drives the demo. For each step it sends a `demo_step` event to one robot; the robot speaks it and returns an `ack`. The orchestrator only advances once the previous step's speech has finished, keeping the robots in sync.
- **Generated speech.** Every step uses `generate=True`: the `text` field is an instruction, and the robot's own LLM/persona generates the actual spoken line — building natural conversation history across the demo rather than reciting fixed scripts.
- **Emotion tags.** Lines begin with a tag such as `[GREETING]`, `[WAVE]`, `[POINT]` or `[HAPPY]`, which drives gestures/expression on robots that support it.
- **Q&A windows.** After each project the guide opens a Q&A window. Visitors speak to any robot (speech → transcription → LLM/RAG → speech). The window closes when a visitor says “move on”, when the robot naturally wraps up, or when the operator presses “Move On” on the dashboard.

Running the demo

- The operator selects the guide and the ordered list of project robots (robot-selection modal, or `POST /demo/start` with a list of robot ids).
- The script below is the default sequence; a dynamic builder produces the same structure for any chosen set of robots.
- An ad-hoc Q&A window can be opened at any time via `POST /demo/qa`.

2 · Demo Script

The exact sequence, generated from `server/demo/demo_script.py`. Each step's guide robot or project robot receives the instruction shown; its LLM turns that instruction into the spoken line. “Q&A” marks a step that opens a question window (manual advance).

#	Step ID	Robot	Instruction to the robot	Notes
1	greeting	Pepper (guide)	You are opening the CARES lab demonstration for a group of visitors. Welcome them warmly, introduce yourself as Pepper the lab guide, and let them know you are excited to show them around today. Keep it to 2 sentences. Start with [GREETING].	gen · 60s
2	lab_intro	Pepper (guide)	Briefly explain what CARES stands for (Centre for Automation and Robotic Engineering Science) and what the lab researches — intelligent robots that can communicate, collaborate, and assist people in real-world environments. Keep it to 2-3 sentences. Start with [WAVE].	gen · 60s
3	overview	Pepper (guide)	Set expectations for the demo: visitors will meet three research robots today, each working on a different project. After each robot speaks, there will be time to ask questions. Keep it to 2 sentences. Use [DEFAULT].	gen · 60s
4	intro_project_a	Pepper (guide)	Introduce the first research project: conversational AI and retrieval-augmented generation. Explain that this project focuses on how robots can hold long, contextually aware conversations by combining language models with a knowledge base. Keep it to 2 sentences. Use [DEFAULT].	gen · 60s
5	introduce_chat_box	Pepper (guide)	Hand off to ChatBox, the robot leading the conversational AI research. Point towards ChatBox and invite them to say hello to the visitors. 1-2 sentences. Use [POINT].	gen · 50s
6	chatbox_greeting	ChatBox	Greet the visitors warmly for the first time. You are ChatBox. Introduce yourself and express genuine excitement about meeting the visitors. 2 sentences. Start with [WAVE].	gen · 50s
7	chatbox_prompt	Pepper (guide)	Ask ChatBox to explain their research project to the visitors. 1 sentence. Use [DEFAULT].	gen · 40s
8	chatbox_project	ChatBox	Explain your research on retrieval-augmented generation (RAG) to a non-expert audience. Cover: what RAG is, how combining language models with a searchable knowledge base helps robots give accurate answers, and how you maintain context across a long conversation. Make it engaging and accessible. 3-4 sentences. Use [DEFAULT].	gen · 90s
9	qa_invite_a	Pepper (guide)	Open a Q&A; session after ChatBox's presentation. Invite visitors to ask questions — they can speak directly to ChatBox or to you. Let them know you will wait until everyone is ready to move on. 2 sentences. Use [DEFAULT].	Q&A window · gen · 60s
10	transition_to_b	Pepper (guide)	Transition from the ChatBox Q&A; to the second project. Give a brief, warm sign-off to ChatBox and announce you are moving on. 1-2 sentences. Use [DEFAULT]. Include 'let us move on' in your response.	gen · 50s
11	intro_project_b	Pepper (guide)	Introduce the second research project: emotion-aware interaction. Explain that this project studies how robots can recognise a person's emotional state and adapt their communication style accordingly. 2 sentences. Use [DEFAULT].	gen · 60s

#	Step ID	Robot	Instruction to the robot	Notes
12	introduce_navel	Pepper (guide)	Hand off to Navel, the robot leading emotion-aware interaction research. Point towards Navel and invite them to say hi to everyone. 1-2 sentences. Use [POINT].	gen · 50s
13	navel_greeting	Navel	Greet the visitors warmly for the first time. You are Navel. Introduce yourself and mention that meeting new people is literally part of your research. 2 sentences. Start with [WAVE].	gen · 50s
14	navel_prompt	Pepper (guide)	Ask Navel to share what their research is about with the visitors. 1 sentence. Use [DEFAULT].	gen · 40s
15	navel_project	Navel	Explain your emotion-aware interaction research to a non-expert audience. Cover: that you detect facial expressions and tone of voice in real time, how you adapt your speaking style based on what you detect (e.g. slower when someone looks confused, warmer when someone seems upset), and your goal of making conversation with a robot feel natural. 3-4 sentences. Use [DEFAULT].	gen · 90s
16	qa_invite_b	Pepper (guide)	Open a Q&A; session after Navel's presentation on emotion-aware interaction. Invite visitors to ask questions — they can speak to Navel or to you. Let them know you will wait until everyone is ready to continue. 2 sentences. Use [DEFAULT].	Q&A window · gen · 60s
17	transition_to_c	Pepper (guide)	Transition from the Navel Q&A; to the third and final project. Brief warm sign-off to Navel and announce the move to the next project. 1-2 sentences. Use [DEFAULT]. Include 'let us move on' in your response.	gen · 50s
18	intro_project_c	Pepper (guide)	Introduce the third research project: human-aware navigation. Frame it as the question: how can a robot move through a crowded space safely, politely, and predictably — the way a person would? 2 sentences. Use [DEFAULT].	gen · 60s
19	introduce_silbot	Pepper (guide)	Hand off to Silbot, the robot specialising in human-aware navigation. Point towards Silbot and invite them to come say hello to the visitors. 1-2 sentences. Use [POINT].	gen · 50s
20	silbot_greeting	Silbot	Greet the visitors warmly for the first time. You are Silbot. Introduce yourself and briefly mention that you navigate spaces with awareness and courtesy. 2 sentences. Start with [WAVE].	gen · 50s
21	silbot_prompt	Pepper (guide)	Ask Silbot to explain their navigation research to the visitors. 1 sentence. Use [DEFAULT].	gen · 40s
22	silbot_project	Silbot	Explain your human-aware navigation research to a non-expert audience. Cover: that rather than just avoiding obstacles, you predict where people are moving, plan routes that do not cut through conversations or crowd groups, that you were trained in simulation and tested in real office corridors at CARES, and your goal of moving through shared spaces the way a respectful colleague would. 3-4 sentences. Use [DEFAULT].	gen · 90s
23	qa_invite_c	Pepper (guide)	Open a Q&A; session after Silbot's presentation on human-aware navigation. Invite visitors to ask questions about navigation and social robotics — they can speak to Silbot or to you. Let them know you will be here until everyone is ready to wrap up. 2 sentences. Use [DEFAULT].	Q&A window · gen · 60s
24	wrap_up	Pepper (guide)	Close the main part of the demo. Thank ChatBox, Navel, and Silbot by name. Summarise that each robot brings a unique capability and together you are all working towards robots that can truly work alongside people. 2-3 sentences. Use [HAPPY].	gen · 60s

#	Step ID	Robot	Instruction to the robot	Notes
25	open_floor	Pepper (guide)	Open a general Q&A; — the demo is complete but the floor is open. Invite visitors to approach any of the robots or speak to you with any remaining questions. Thank them warmly for joining the CARES lab demonstration today. 2-3 sentences. Use [DEFAULT].	Q&A window · gen · 60s

Legend — *gen*: line generated by the robot's LLM from the instruction; *verbatim*: spoken as written; the seconds value is the per-step timeout; shaded rows open a Q&A window (operator advances with “Move On”).